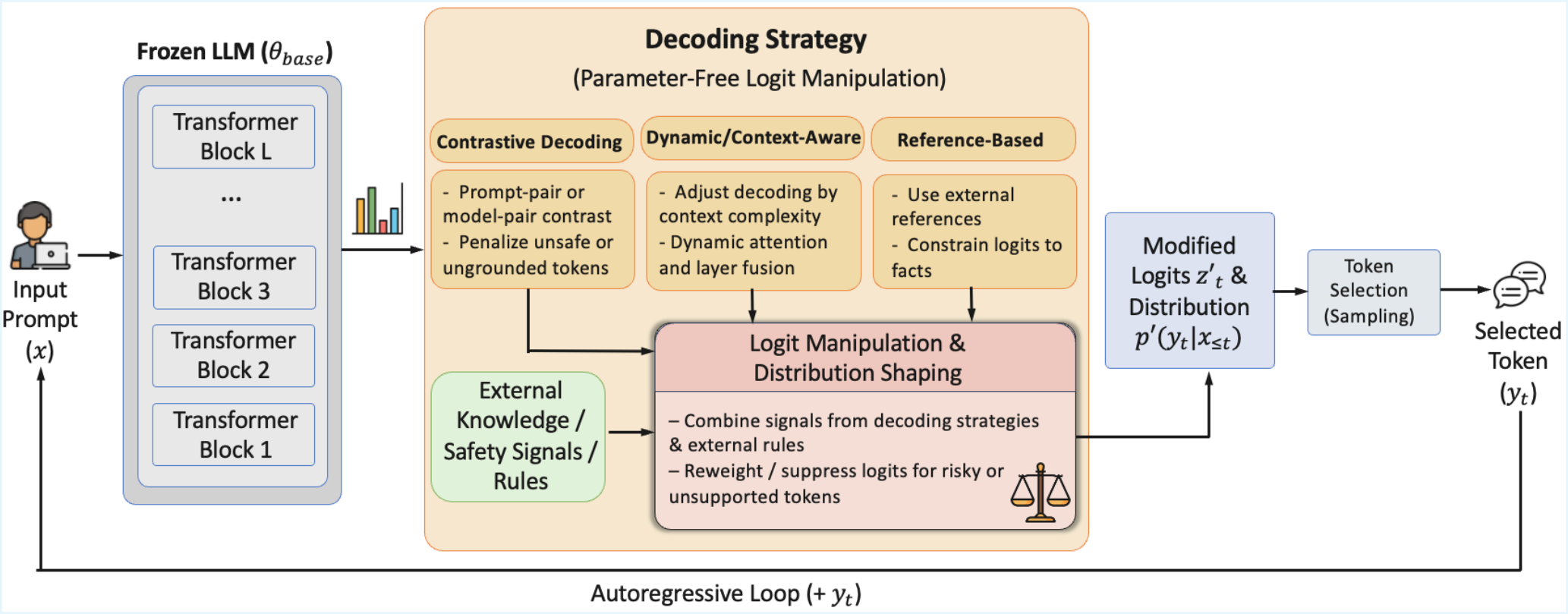

Frozen LLM (θ_{base})
Transformer Block L
...
Transformer Block 3
Transformer Block 2
Transformer Block 1
Input Prompt (x)
Decoding Strategy
(Parameter-Free Logit Manipulation)
Contrastive Decoding
Dynamic/Context-Aware
Reference-Based
- Prompt-pair or model-pair contrast
- Penalize unsafe or ungrounded tokens
- Adjust decoding by context complexity
- Dynamic attention and layer fusion
- Use external references
- Constrain logits to facts
External Knowledge / Safety Signals / Rules
Logit Manipulation & Distribution Shaping
– Combine signals from decoding strategies & external rules
– Reweight / suppress logits for risky or unsupported tokens
Modified Logits z'_t & Distribution $p'(y_t|x_{\leq t})$
Token Selection (Sampling)
Selected Token (y_t)
Autoregressive Loop (+ y_t)